Principle:

To study about the chromosomal aberration which can occur due to ionizing radiations or chemicals. On the basis of breaks and reunions in the chromosomal segment different types of aberrations can be recognized.

Requirements:

Copper wire, Alphabets marked (A to H) yellow colour beads denotes gene, and red colour bead without alphabet denote centromere. Using this materials make different kinds of chromosomal segments with specific gene sequence, that can be given to the students and asked to analyse the aberration involved in it.

Procedure:

1. Make a normal chromosome model using copper wire and yellow beads and place it on the table. In the model chromosome with gene sequence A to H, along with centromere (red bead).
2. For Deletion - Give yellow colour beads without one or more marked alphabets A to H (The lack of any one or more beads denotes deletion type of chromosomal aberration).
3. For Duplication – Give yellow colour beads with addition of one or more marked alphabets A to H (The repetition of one or more beads denotes duplication type of chromosomal aberration).
4. For Inversion – Give yellow colour beads which marked alphabets from A to H as in normal chromosome. (There is no addition or deletion of beads (A to H) given, so the students can construct the inverted segment of the chromosome using the given beads).

   Based on the type of beads given the student has to identify and construct the relevant chromosomal aberration.

17 A. Chromosomal Aberration – Deletion

Reasons:

1. The deletion of the chromosomal segement A and D. (Refer figure 17a)
2. When there is a loss of a segment of the genetic material in a chromosome it is called deletion.

Significance:

Most of the deletions lead to death of an organism.

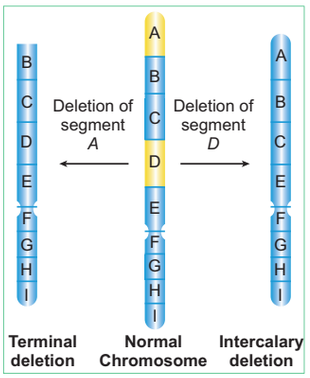


Figure: 17 a: Deletion

17 B. Chromosomal Aberration - Duplication

Reasons:

1. When a segment of a chromosome is present more than once in chromosome, then it is called duplication (Tandem duplication)
2. The order of the genes in a chromosome is A, B, C, D, E, F, G, H and Due to aberration, the genes B and C are duplicated and the sequence genes becomes A, B, C, B, C, D, E, F, G, H and I. (Refer figure 17b)

Significance:

Some duplications are useful in the evolution of the organism.

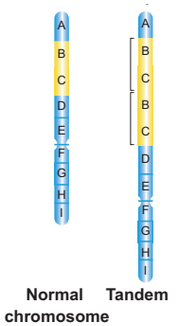


Figure 17 b: Duplication

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